

VILA REAL AP, Portugal

WMO: 085670

Lat: 41.2742N

Lon: 7.7172W

Elev: 562

StdP: 94.76

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.1	0.0	-6.7	2.3	3.2	-4.8	2.7	3.9	8.4	8.2	7.1	8.8	1.1	30	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.5	33.5	19.6	31.7	19.1	30.0	18.6	21.0	30.6	20.2	29.4	19.4	28.1	2.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	13.6	24.5	16.9	12.9	23.4	16.2	12.3	22.6	63.1	30.7	60.2	29.6	57.4	28.3	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.3	5.3	4.5	DB	-3.1	36.4	1.2	1.3	-3.9	37.3	-4.6	38.1	-5.3	38.8	-6.1	39.7
			WB	-4.2	22.6	1.2	0.9	-5.0	23.3	-5.7	23.8	-6.4	24.3	-7.2	25.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	6.5	7.8	10.2	11.9	15.2	19.0	21.3	21.8	19.3	14.8	9.3	6.8	
	DBStd	6.40	2.84	2.75	3.19	3.79	3.87	3.97	3.65	3.49	3.32	3.30	2.96	2.78	
	HDD10.0	398	112	71	37	21	3	0	0	0	0	3	47	105	
	HDD18.3	2053	366	296	251	197	112	39	12	7	27	117	270	358	
	CDD10.0	1752	3	9	44	77	165	271	351	366	281	153	27	5	
	CDD18.3	364	0	0	0	3	16	61	105	114	57	8	0	0	
	CDH23.3	3689	0	0	1	32	176	637	1118	1175	492	59	0	0	
	CDH26.7	1411	0	0	0	2	41	231	471	501	154	11	0	0	
Wind	WSAvg	1.7	1.6	1.8	1.9	1.9	1.9	1.9	2.0	1.8	1.5	1.4	1.6	1.4	
Precipitation	PrecAvg	1112	160	170	97	90	70	53	15	16	49	108	125	160	
	PrecMax	1812	444	449	357	263	190	238	74	68	162	275	459	619	
	PrecMin	615	0	23	0	4	10	3	0	0	2	1	0	9	
	PrecStd	294	112	127	88	66	45	53	20	18	40	79	99	143	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	18.0	22.5	26.3	30.6	33.8	35.3	35.9	33.2	28.3	18.6	14.3	
		MCWB	10.9	11.0	12.4	15.6	18.3	20.3	19.9	20.3	18.5	15.9	13.7	11.8	
	2%	DB	12.7	15.6	20.4	23.9	27.4	31.3	33.4	33.5	30.4	24.7	16.4	13.0	
		MCWB	10.6	10.6	11.8	14.6	17.2	19.2	19.7	19.3	18.0	15.8	12.9	11.3	
	5%	DB	11.7	13.8	18.3	21.4	25.2	29.5	31.4	31.6	28.3	22.2	14.9	12.0	
		MCWB	10.1	9.5	11.2	13.6	16.1	18.7	19.4	18.9	17.3	15.1	12.2	10.6	
	10%	DB	10.8	12.2	15.9	18.7	22.9	27.2	29.3	29.6	26.3	20.1	13.7	11.0	
		MCWB	9.3	8.9	10.2	12.4	15.4	17.9	18.9	18.4	16.8	14.6	11.6	9.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.4	12.7	14.0	16.8	20.0	21.7	21.9	22.1	20.1	18.5	15.6	12.8
			MCDB	13.2	15.4	19.8	24.6	27.6	31.1	31.6	32.6	29.7	22.7	17.3	13.5
2%		WB	11.4	11.5	12.8	15.2	18.3	20.5	20.9	20.7	19.0	17.2	14.1	11.8	
		MCDB	12.1	14.1	17.4	22.3	25.5	29.4	30.7	30.5	27.8	21.5	15.3	12.6	
5%		WB	10.5	10.4	12.0	14.1	16.9	19.5	20.1	19.8	18.2	16.3	13.0	10.8	
		MCDB	11.4	12.9	16.2	20.3	23.6	27.8	29.4	29.0	26.2	20.2	14.3	11.8	
10%		WB	9.5	9.5	11.1	12.8	15.7	18.5	19.2	19.0	17.4	15.4	11.8	9.8	
		MCDB	10.6	11.8	14.9	17.8	21.7	26.1	28.1	27.8	24.5	18.7	13.2	10.9	
Mean Daily Temperature Range	5% DB	MDBR	6.3	8.1	9.2	9.6	11.1	12.3	13.4	13.5	11.8	8.9	6.7	6.1	
		MCDBR	7.1	10.5	12.8	13.6	14.3	15.0	15.5	15.4	14.4	12.2	8.0	6.4	
		MCWBR	5.2	6.5	7.1	6.9	7.2	7.1	6.7	6.4	6.1	6.0	5.3	4.9	
	5% WB	MCDBR	5.6	8.9	10.5	12.7	13.4	14.3	14.9	14.6	13.0	9.4	6.3	5.7	
		MCWBR	4.6	6.1	6.3	6.6	7.0	7.1	6.8	6.6	5.9	5.2	4.8	4.7	
Clear-Sky Solar Irradiance	taub	0.299	0.310	0.341	0.345	0.353	0.356	0.344	0.356	0.344	0.335	0.308	0.296		
	taud	2.546	2.504	2.431	2.423	2.422	2.428	2.445	2.403	2.460	2.510	2.547	2.550		
	Ebn at Noon	866	906	910	926	921	916	924	904	894	864	846	839		
	Edh at Noon	71	85	103	111	114	114	111	112	99	84	70	65		
All-Sky Solar Radiation	RadAvg	1.80	2.84	3.92	5.05	6.27	7.13	7.44	6.63	5.07	3.18	2.04	1.61		
	RadStd	0.29	0.42	0.62	0.63	0.65	0.45	0.32	0.34	0.37	0.41	0.39	0.24		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.60	N/A	N/A	N/A	N/A	N/A	N/A	-151	+180	N/A		
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page